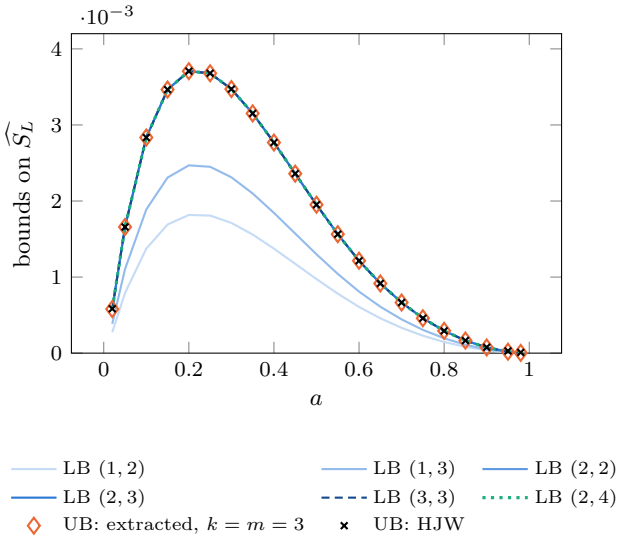


Figure 1: Convex roof of the linear entropy $S_L = 1 - \operatorname{tr} \rho_A^2$ for the Horodecki 2×4 family ρ_a. Lines: lower bounds at the levels (k, m) of the simplified hierarchy; from $k = 2$ on they coincide, including $(2, 4)$ (green, dotted). Markers: upper bounds from the decomposition extracted from the SDP solution at $k = m = 3$ (orange diamonds) and from HJW gradient descent (black crosses).

1 Extraction

$$x \tag{1}$$

$$y \tag{2}$$

Theorem 1 *z*

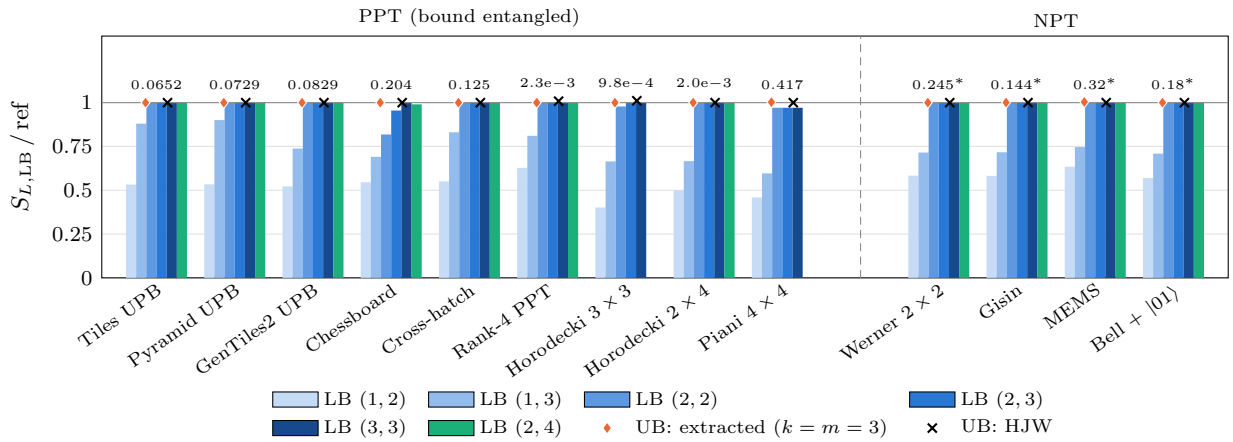


Figure 2: Convex roof of the linear entropy $S_L = 1 - \mathrm{tr}\,\rho_A^2$ for well-known bound entangled (PPT) and NPT states. Bars: lower bounds from the simplified hierarchy at the levels (k, m) shown in the legend (blue, light to dark, up to $k = m = 3$; green: $(2, 4)$, for comparison with $(3, 3)$). Markers: upper bound from the decomposition extracted from the SDP solution at $k = m = 3$ (orange diamonds), upper bound from HJW gradient descent over decompositions (black crosses), drawn left and right of each group; values above the axis are clipped and printed. Every value is divided by the reference printed above each group: the exact roof for two qubits (marked *), the best upper bound otherwise. The level $(2, 4)$ is missing for Horodecki 3×3, Piani 4×4, where the SDP exceeds the memory of our machine (more than 32 GB). Members of parametric families: GenTiles2 UPB (3×4); Chessboard (extremal member); Cross-hatch (grid state); Rank-4 PPT (Nat. Commun. 5, 6297 (2014)); Horodecki 3×3 ($a = 0.5$); Horodecki 2×4 ($b = 0.5$); Werner 2×2 (singlet weight $p = 0.8$); Gisin ($\lambda = 0.9$, $\theta = \pi/8$); MEMS ($C = 0.8$); Bell + $|01\rangle$ ($p = 0.6$).

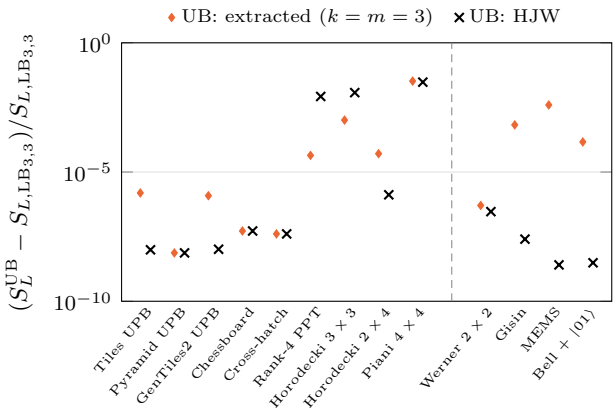


Figure 3: Relative gap between the upper bounds on $\widehat{S_L}$ and the SDP lower bound at $k = m = 3$, for the states of Fig. 2 (PPT left of the dashed line, NPT right). Orange: decomposition extracted from the SDP solution (Sec. 1); black: HJW gradient descent from random starts. Points at 10^{-10} are gaps below solver precision.

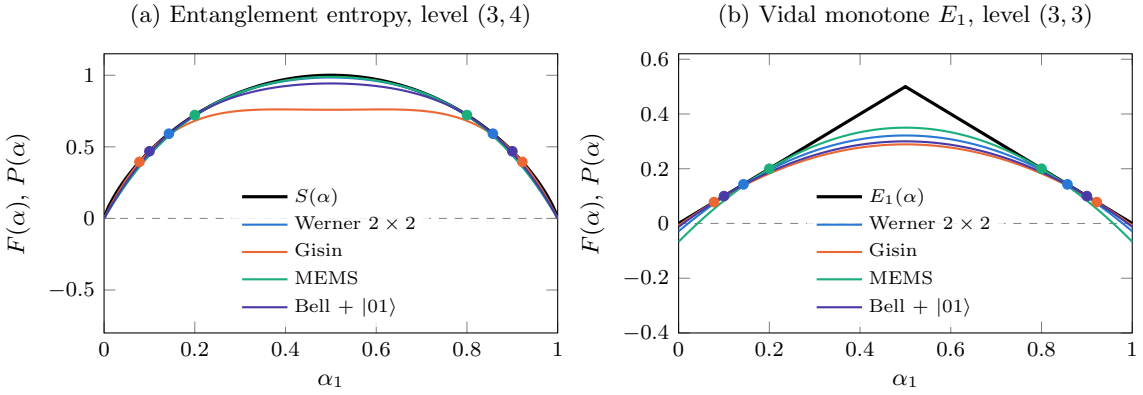


Figure 4: Polynomials $P \leq F$ selected by the LP for four two-qubit states (Werner 2×2: singlet weight $p = 0.8$; Gisin: $\lambda = 0.9$, $\theta = \pi/8$; MEMS: $C = 0.8$; Bell + $|01\rangle$: $p = 0.6$), as functions of the Schmidt coefficient α_1; the spectrum is $(\alpha_1, 1 - \alpha_1)$, so both F and P are symmetric under $\alpha_1 \to 1 - \alpha_1$. (a) Entanglement entropy at level $(k, m) = (3, 4)$. (b) Vidal monotone $E_1 = \min(\alpha_1, 1 - \alpha_1)$ at level $(3, 3)$. Dots: the Schmidt coefficients $(1 \pm \sqrt{1 - C^2})/2$ shared by all states of the optimal (Wootters) decomposition, where P touches F when the bound is tight.

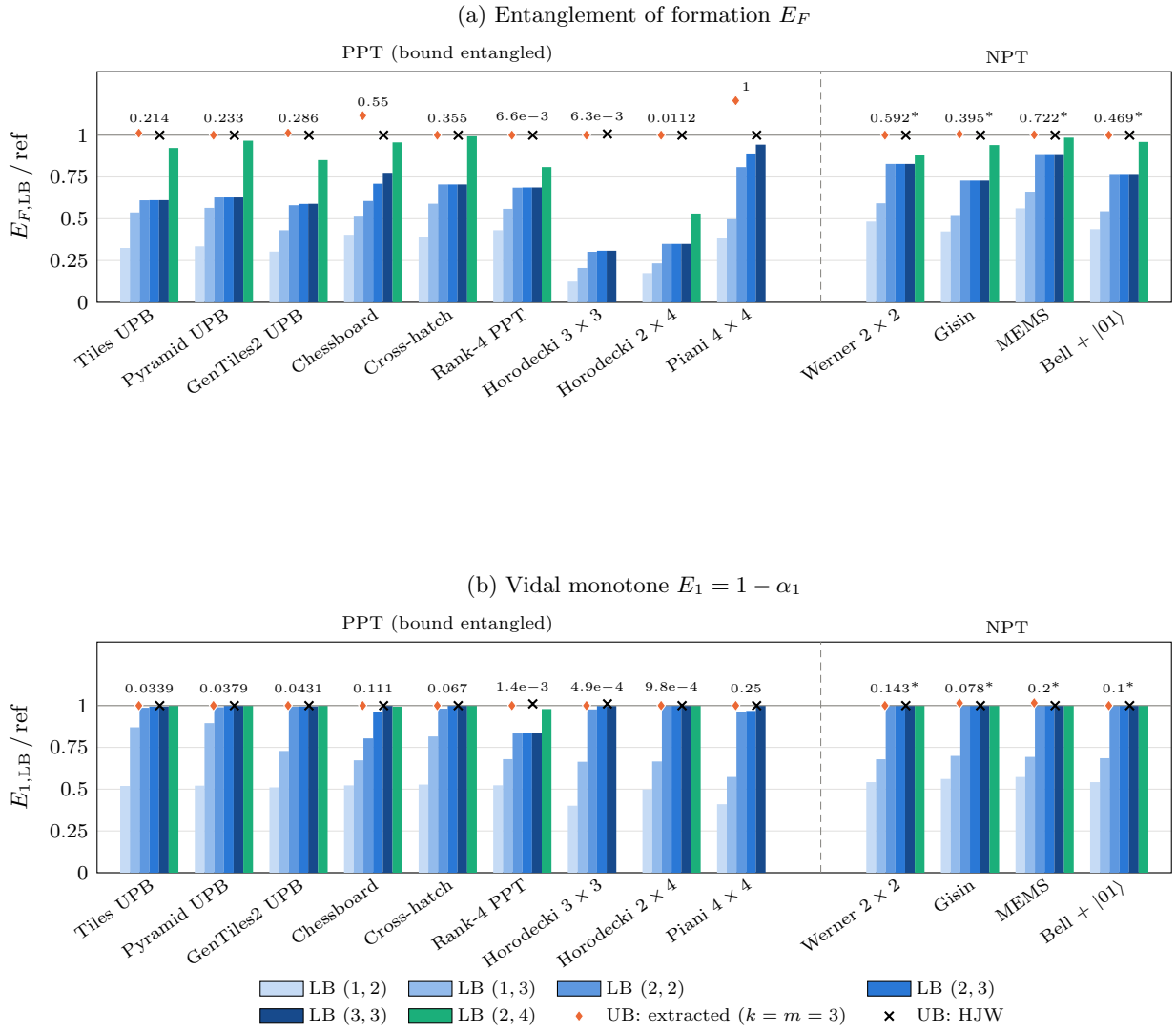


Figure 5: Lower bounds on (a) the entanglement of formation and (b) the convex roof of the Vidal monotone E_1 from the SDP with embedded LP (polynomials of degree m), for the states of Fig. 2. Bars: lower bounds from the simplified hierarchy at the levels (k, m) shown in the legend (blue, light to dark, up to $k = m = 3$; green: $(2, 4)$, for comparison with $(3, 3)$). Markers: upper bound from the decomposition extracted from the SDP solution at $k = m = 3$ (orange diamonds), upper bound from HJW gradient descent over decompositions (black crosses), drawn left and right of each group; values above the axis are clipped and printed. Every value is divided by the reference printed above each group: the exact roof for two qubits (marked *), the best upper bound otherwise. The level $(2, 4)$ is missing for Horodecki 3×3, Piani 4×4, where the SDP exceeds the memory of our machine (more than 32 GB). The decomposition giving the orange markers is extracted from the SDP of the optimal polynomial $P^* \leq F$ found at $k = m = 3$. For two qubits, odd m gives the same polynomials as $m - 1$.

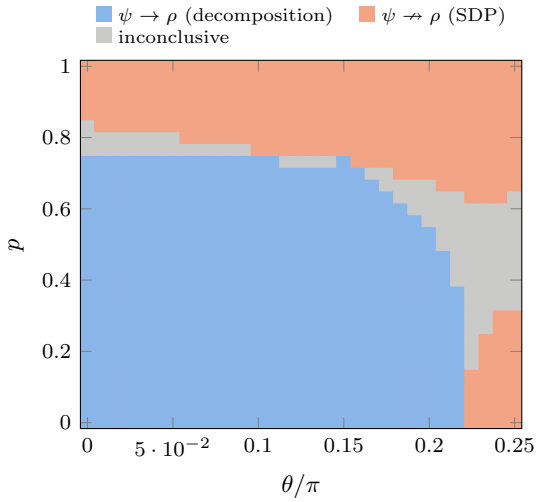


Figure 6: LOCC convertibility from the pure state $|\psi\rangle$ with entanglement spectrum $(0.6, 0.3, 0.1)$ to the two-qutrit states $\rho(p,\theta) = p\,|\Phi_3\rangle\langle\Phi_3| + (1-p)\,|\chi_\theta\rangle\langle\chi_\theta|$, with $|\Phi_3\rangle$ maximally entangled and $|\chi_\theta\rangle = \cos\theta\,|00\rangle + \sin\theta\,|11\rangle$, from the SDP of Eq. (**??**) at level $(k,m) = (3,4)$. Blue: the decomposition extracted from the optimal $\Phi_{k,m}$ satisfies Eq. (1), so $\psi \to \rho$ [Theorem **??**(ii)]. Orange: $s_{k,m}$ exceeds the grid correction, so $\psi \nrightarrow \rho$ [Theorem **??**(i)]. Gray: neither certificate applies at this level.

			simplified, $(k,m)=(2,3)$			full, $(k,m)=(3,3)$			simplified, $(k,m)=(2,4)$		
State	$d_A \times d_B$	r	raw	reduced	time	raw	reduced	time	raw	reduced	time
Werner	2×2	4	32	20	5 s	64	20	< 1 s	64	40	9 s
Horodecki	2×4	5	128	30	2 s	512	35	2 s	256	60	16 s
Tiles UPB	3×3	4	243	30	5 s	729	20	5 s	729	90	226 s
Horodecki	3×3	7	243	84	13 min	729	84	40 s	729	252	OOM
Isotropic	3×3	9	243	135	586 s	729	165	568 s	729	405	–
GenTiles2 UPB	3×4	5	432	45	31 s	1 728	35	2 s	1 296	135	31 min
Piani	4×4	6	1 024	84	289 s	4 096	56	11 s	4 096	336	OOM

Table 1: Side length of the SDP variable and solve time of the bound on the convex roof of the linear entropy, for representative states of each dimension, at the levels $(2,3)$ and $(2,4)$ of the simplified relaxation [Eq. (**??**)] and at $k = m = 3$, where it is the full SDP of Eq. (**??**). Raw: $D^k d_A^{m-k}$, the size of $\Phi_{k,m}$ without reductions ($D = d_A d_B$). Reduced: $\binom{k+r-1}{k} d_A^{m-k}$, after restricting the k full copies to the symmetric subspace of the support of ρ, with $r = \operatorname{rank} \rho$ (Appendix **??**); in addition, the positivity constraints are split into Schur blocks and the marginal copies $A_{k+1}, \ldots, A_m$ are parametrized by their permutation invariance. At equal reduced size, levels with marginal copies ($m > k$) are slower, because of their additional partial-transpose constraints and permutation invariance equalities. Times: interior-point solver QICS on a 16-core laptop with 23 GB of memory, measured with 13 runs in parallel (one thread each), so they are indicative only. OOM: the solver needs more than the 23 GB of memory; –: not attempted.